GRADE 7 ACCELERATED · UNIT 4 · LESSON 4

Real-World Problems and Significant Digits

Independent Set 4.04 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. How many significant digits are in 6,030 grams?

3

The middle zero counts; the trailing zero does not, because no decimal point is written.

2. How many significant digits are in 0.00710 meters?

3

Leading zeros never count; the trailing zero does, because a decimal point is written.

3. A rectangle measures 2.4×10^2 m by 3.15×10^2 m. Find its area, to the correct number of significant digits.

7.6×10^4 ***square meters***

75,600 exactly, kept to two significant digits.

4. $(6.30 \times 10^4) + (2.8 \times 10^3)$ meters, both measured. Write the sum with the correct precision.

6.58×10^4 ***m***

63,000 and 2,800 both stop at the hundreds place, so the sum, 65,800, does too.

5. A car travels 1.80×10^2 km in 4.0 hours. Find its average speed, to the correct number of significant digits.

4.5×10^1 ***km/h***

45 exactly; two significant digits.

6. $(3.7 \times 10^{-2}) - (8.45 \times 10^{-4})$ grams, both measured. Write the difference with the correct precision.

3.6×10^{-2} ***g***

0.036155 exactly; 0.037 stops at the thousandths place, so the answer stops there.